GlycoTwin architecture: simulate, train, evaluate, serve

Boxes are files in the repository; arrows show data flow. White = script, tinted = data or model file, navy = clinician. Slide 2: care pathway.

1 OFFLINE: SIMULATE, TRAIN, EVALUATE

generate_datasets.py
data/ · seeded simulator, illustrative parameters

synthetic_ehr.csv
200 personas: labs, meds, history, TCF7L2

synthetic_wearables.csv
132,800 windows of 15 min

train_fusion_model.py
- 34 features: 18 wearable + 16 EHR, EHR joined to every window
- patient split 150 / 50
- HGB classifier + regressor
- alert thresholds 0.51 / 0.52 by F2 on grouped CV of training patients

glycotwin_model.joblib
classifier, regressor, thresholds, feature list, category levels, split

evaluate.py
held-out, CV, baselines, ablation, latency

metrics.json
the only source of performance numbers

bundle + metrics.json →

2 SERVING: TWO INFERENCE PATHS, DOCTOR IN THE LOOP

Doctor in the loop
- Diabetologist or diabetes educator reviews the risk board, alerts and model card
- Decides the outreach: a call, a medication review
- Decision support only; no automated dosing

(a) dashboard/app.py · Streamlit
live inference on the selected window (about 15 ms on CPU); Data Ingest runs 5 checks on uploads

(b) build_standalone_dashboard.py
scores the demo patients in Python at build time and embeds the outputs

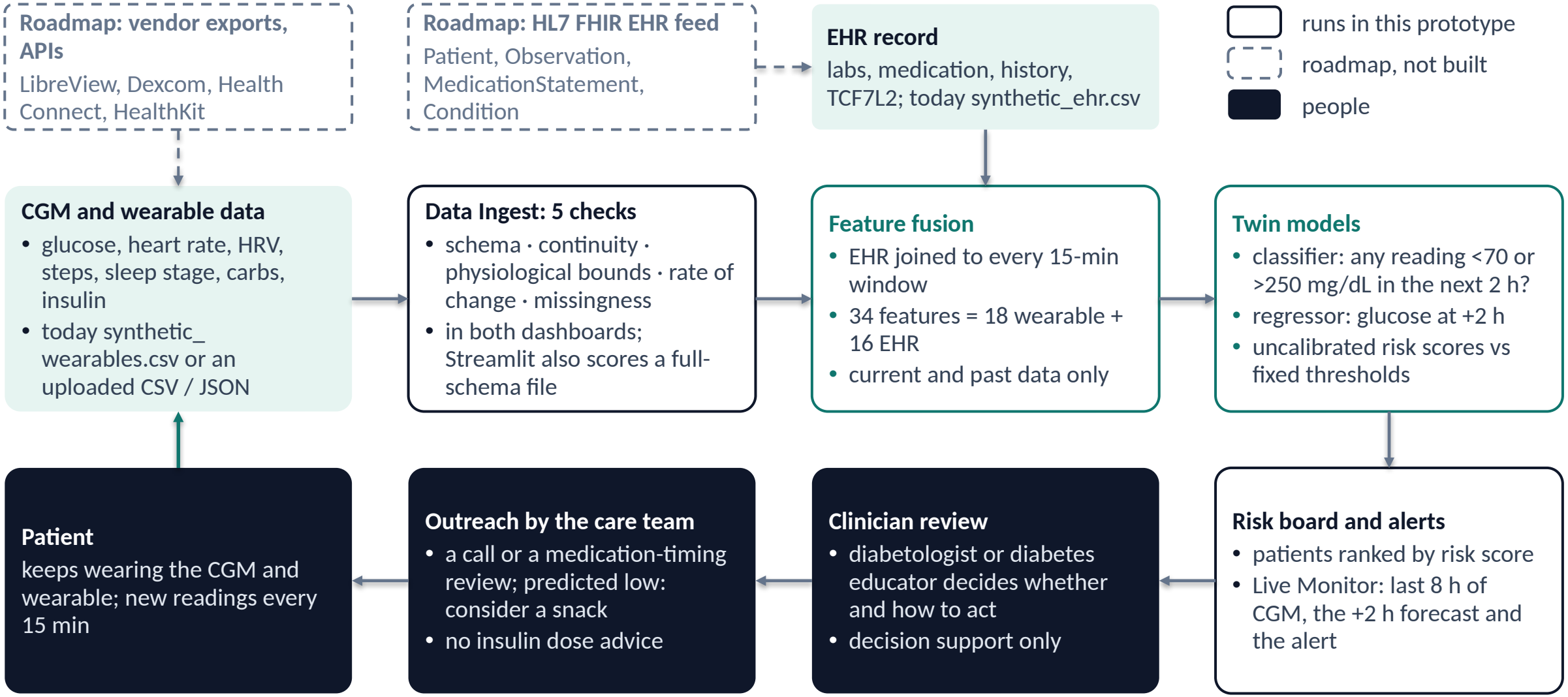
dashboard/index.html + plotly.min.js
static replay page; no model runs in the browser; uploads get the 5 ingest checks

public/ on Firebase Hosting
glycotwin-app.web.app, opens on a laptop or a phone

Synthetic data only · research prototype, not a medical device · not for dosing decisions

GlycoTwin in the care pathway: from device data to clinician action

Conceptual view of slide 1. Solid boxes run in this prototype today, on synthetic data; dashed boxes are roadmap, not built; navy boxes are people.



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